

Question

In the experiment, the relative intensities of three different mass signals of dichlorodibromomethane in the mass spectrum were measured as $M(240) : 38.67\%$, $M + 2 : 100\%$, $M + 4 : 88.64\%$.

It is known that the main isotopes of chlorine are ^{35}Cl and ^{37}Cl with an abundance ratio of approximately 3 : 1, and the main isotopes of bromine are ^{79}Br and ^{81}Br with an abundance ratio of approximately 1 : 1.

Which of the following statements is correct (a calculation error of less than 0.5% is considered acceptable):

- A.** All other options are incorrect
- B.** The abundance of ^{35}Cl is 74.2%
- C.** The abundance of ^{37}Cl is 25.2%
- D.** The abundance of ^{79}Br is 51.6%.
- E.** The abundance of ^{81}Br is 49.7%.
- F.** The relative signal intensity of $M + 6$ is 34.5%.
- G.** The relative signal intensity of $M + 8$ is 1.57%.
- H.** Let the abundance of ^{13}C be 1%, then the relative signal intensity of $M + 1$ theoretically detected in this experiment is approximately 0.4%.

Answer

Correct Answer: **H**

Detailed Explanation

Let ^{35}Cl be event A with probability x ; ^{79}Br be event B with probability y .

$$P(M) = P(AABB) = xxyy$$

$$P(M+2) = P(\bar{A}ABB + A\bar{A}BB + A\bar{A}B\bar{B} + AAB\bar{B}) = 2x(1-x)yy + 2xxy(1-y)$$

$$P(M+4) = P(\bar{A}\bar{A}BB + \bar{A}A\bar{B}B + \bar{A}AB\bar{B} + A\bar{A}\bar{B}B + A\bar{A}B\bar{B} + AA\bar{B}\bar{B})$$

$$= (1-x)^2yy + 4x(1-x)y(1-y) + xx(1-y)^2$$

According to the problem,

$$P(M) : P(M+2) : P(M+4) = 0.3867 : 1 : 0.8864$$

Finally, we can solve to obtain $x = 75.8\%$, $y = 50.7\%$.

Therefore, the abundance of ^{35}Cl is 75.8% , and the abundance of ^{37}Cl is 24.2% ;

CHECKPOINT

2 PTS

The abundance of ^{35}Cl is 75.8% , and the abundance of ^{37}Cl is 24.2% ;

The abundance of ^{79}Br is 50.7% , and the abundance of ^{81}Br is 49.3% ;

CHECKPOINT

2 PTS

The abundance of ^{79}Br is 50.7% , and the abundance of ^{81}Br is 49.3% ;

$$P(M + 6) = P(A\bar{A}\bar{B}\bar{B} + \bar{A}\bar{A}B\bar{B} + \bar{A}A\bar{B}\bar{B} + \bar{A}\bar{A}\bar{B}B)$$

$$= 2x(1-x)(1-y)^2 + 2(1-x)^2y(1-y) = 30.23\%$$

CHECKPOINT

1 PTS

$$P(M + 6) = 30.23\%$$

$$P(M + 8) = P(\overline{AABB}) = (1 - x)^2(1 - y)^2 = 1.43\%$$

CHECKPOINT

1 PTS

$$P(M + 8) = 1.43\%$$

Therefore, options B-G are all incorrect.

The ratio of the relative signal intensity of $M + 1$ to that of M is equal to the ratio of the abundances of ^{13}C to ^{12}C :

$$\frac{P(M+1)}{P(M)} = \frac{1\%}{1-1\%} = \frac{x}{38.67\%}$$

Solving gives $x \approx 0.4\%$

CHECKPOINT

1 PTS

$$P(M + 1) = 0.4\%$$

Option H is correct.